

Why Private 5G Services are Needed for Implementing Industry 4.0

5G has arrived! Manufacturers would do well to avail the high bandwidth it brings with it to ensure smoother operations as they embrace Industry 4.0.



With manufacturers opting for Industry 4.0 in order to increase productivity, there is a critical need for reliable connectivity that supports high bandwidth and low latency at the same time.

A Wi-Fi network is most suitable for the indoors but is not scalable to a 10 square kilometre area. To do that we need to install a Wi-Fi modem at every 10 to 20 metres, which is impractical and costly. A cellular network is the most appropriate option here. We already have working 4G networks. These networks are predominantly public networks and not private

networks with an enterprise. Similarly, the cloud is also not physically present within an enterprise.

The location of the data center is uncertain. Even if we are aware of its location, the fact that it is travelling through the web network rather than the public network prevents you from knowing how much latency it will incur. This is why this latency needs to be changed. With the introduction of 5G this can be dealt with easily, as it allows you to create an alternative Wi-Fi network within the enterprise. It ensures low latency and high throughput as well as privacy.

Along with that, you also need to have a miniature version of the cloud, which we call the 'H' cloud. It is more reliable for things like augmented reality (AR) and virtual reality (VR), video analytics, robotic operations and autonomous vehicles, and guarantees low latency. It is also important to note that the cloud is essentially for book-keeping and data management and not for any computation tasks.

Information technology applications such as Zoom, as well as system applications and products (SAP) can be run properly on Wi-Fi but operational applications cannot.